JUSTIN HARRY MARTEL, PhD.

Mathematician, Computations, Physical Systems

Education

Doctor of Philosophy (2013--2019)

University of Toronto, Department of Mathematics and Statistics

Supervisor: Dr. R.J. McCann

Thesis: *Applications of Optimal Transportation to Algebraic Topology: How to Build Spines from Singularity*

Master of Science (2011--2013)

University of British Columbia, Department of Mathematics and Statistics.

Supervisor: Dr. Lior Silberman

Bachelor of Science with Honours (2007--2011)
Specialization in Mathematics

University of Ottawa

Supervisor: Dr. Damien Roy

Skills

_Tireless and Tenacious Researcher.

_Readily analyzes research articles and technical publications in mathematics, probability, stochastics, linear algebra, physics, signal processing, cryptography, machine learning, etc.

_Keen lecturer, able to present and illustrate complex ideas in simple terms.

_Broad mathematical knowledge allowing customization and experimentation with, e.g., nearly all Machine Learning algorithms.

Work Experience

Assistant Operations Manager (06/20--present)

Ottawa Moving and Delivery, 339 Churchill Ave N, Ottawa, Ontario.

Reference: Shane Kennedy (613) 247-7918

Nursery Sales, (03/20--06/20)

Nesbitt's Greenhouse and Nursery, 235 Chemin Eardley, Gatineau, Quebec.

Reference: Sylvia Nesbitt (819) 684-6969

Course Instructor: Calculus I, II, Vector Calculus (09/17--12/19)

University of Toronto, Department of Mathematics and Statistics.

Reference: Jemima Merisca (416) 978-7894

Preprints

1. *How to Build Spines from Singularity* (PhD thesis, www.github.com/jhmartel/thesis)
2. *Dold-Thom, Homology, and Optimal Transport* (in progress www.github.com/jhmartel/DT)
3. *Closing Steinberg Symbols in the Mapping Class Group of Hyperbolic Surfaces and Spines for Teichmueller Space* (in progress at www.github.com/jhmartel/MCG)
4. *Kantorovich Singularities in Alexandrov Spaces: New Proof of Perelman's Soul Theorem* (work in progress at www.github.com/jhmartel/Alexandrov)

Contact:

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www.mathassist.ca

Research Interests:

_Optimal Transportation,
_Computational Topology,
_Singularities,
_Classical Mechanics,
_Electromagnetism and Geometry of Ground States,
_Plasma Models
_Machine-Tool Principles and Physical Economy,
_Differential Equations and Modelling

Technical Skills:

_WOLFRAM Mathematica
_Python
_Latex
_Readily learns new computer languages

Languages:

English ●●●●●
French ●●

Qualities:

Hard worker both physically and mentally,
Always learning and developing new skills,
Diverse interests, widely read.
Seeking challenges, and looking to solve impossible problems.